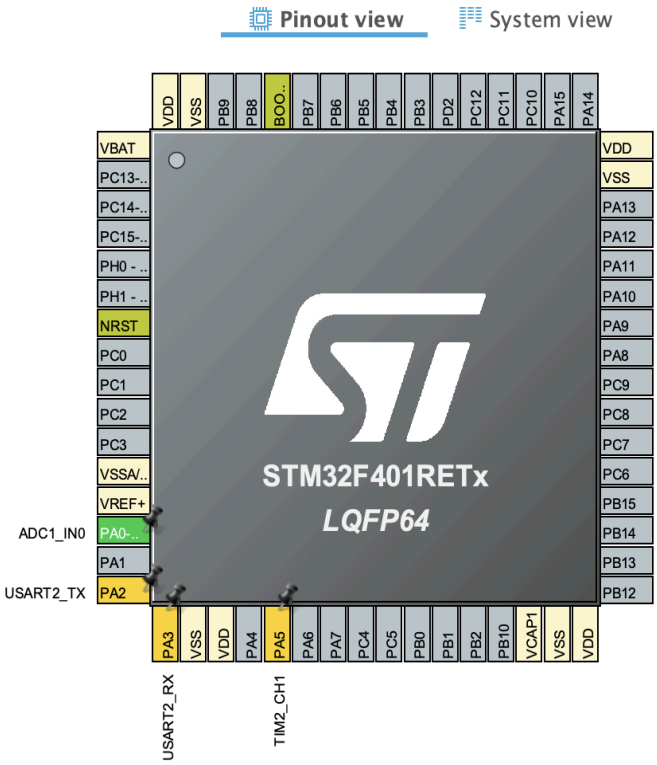
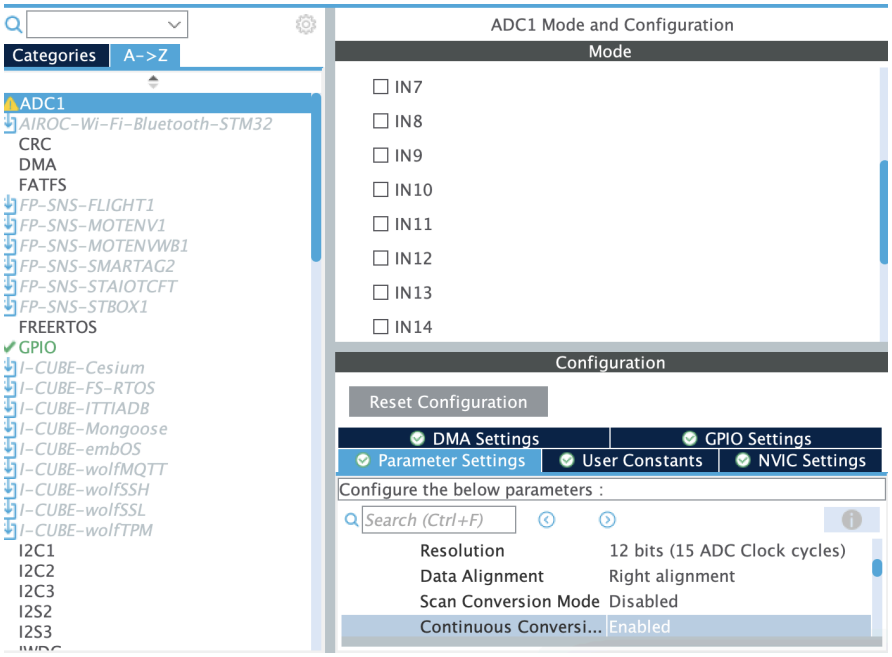
PINOUT CONFIGURATION



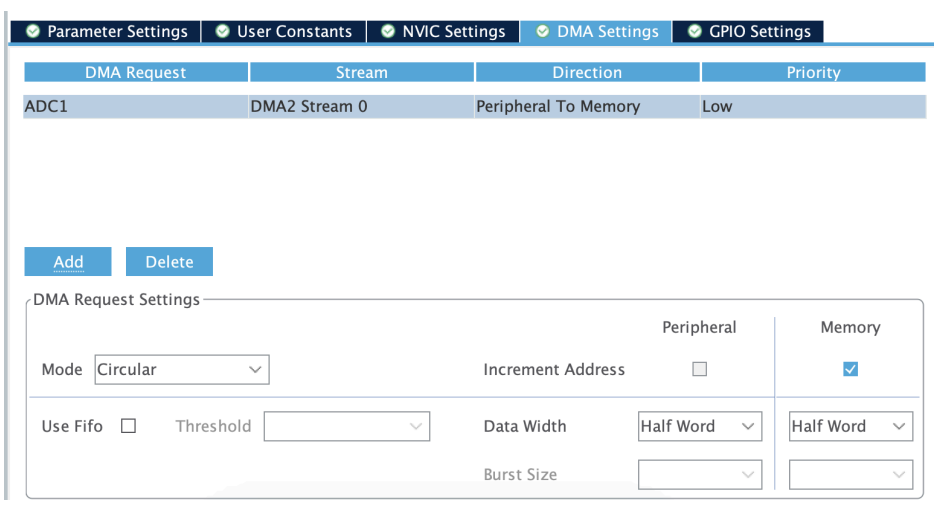
Pinout configuration showing ADC input on PA0, PWM output on PA5, and UART communication via USART2.

ADC CONFIGURATION



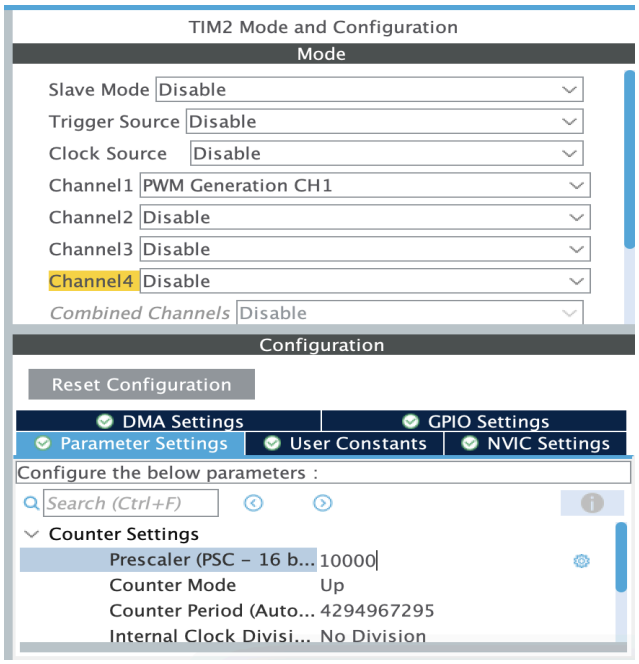
ADC configuration set to 12-bit resolution and continuous conversion mode for real-time signal acquisition.

DMA CONFIGURATION



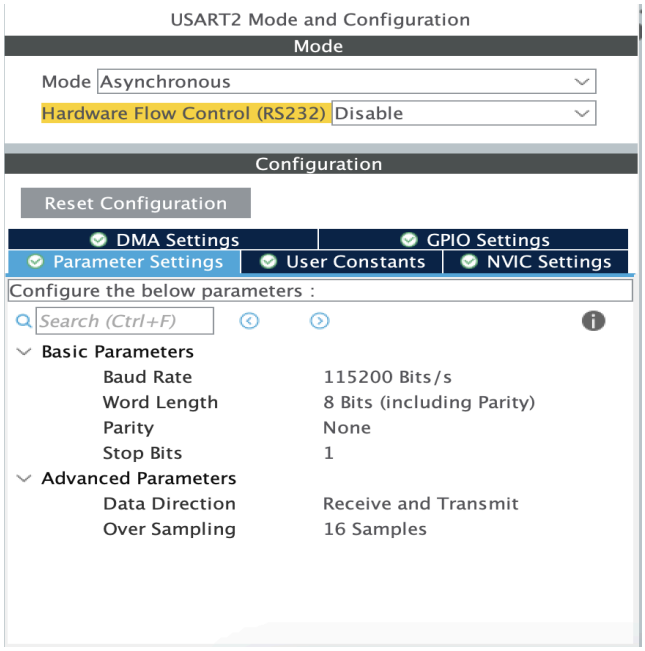
DMA configuration in circular mode for continuous data transfer from ADC to memory without CPU intervention.

PWM CONFIGURATION



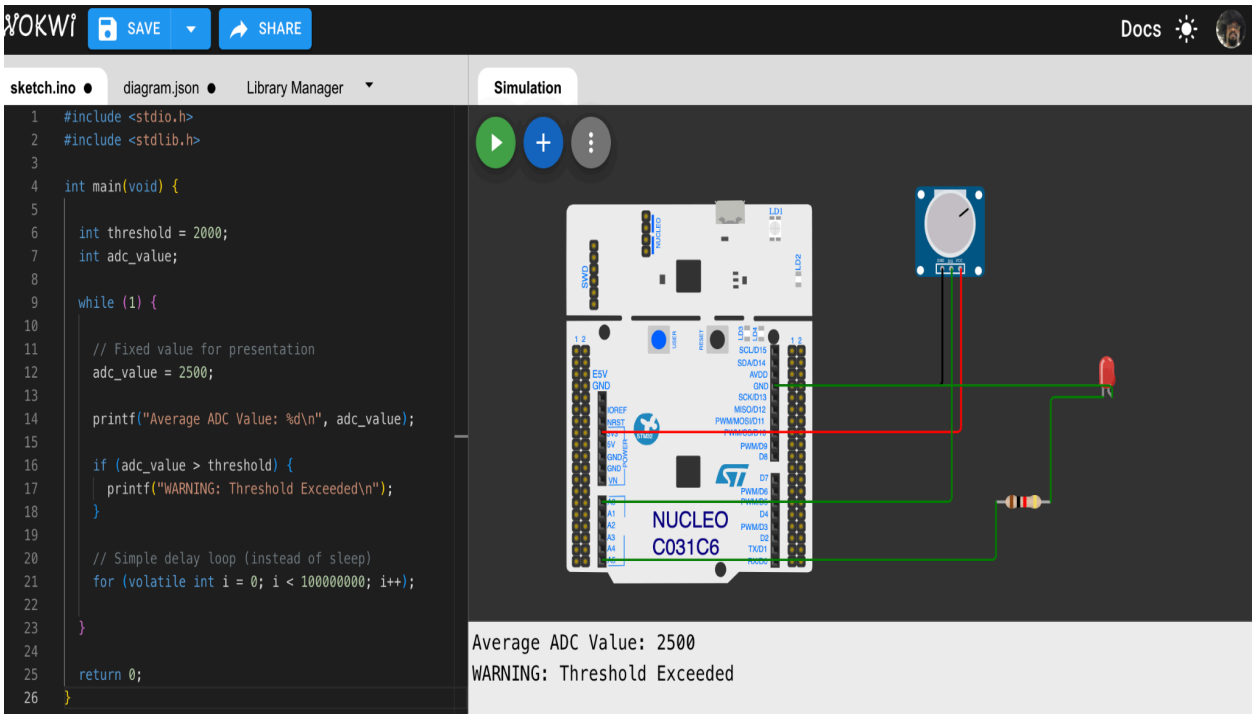
PWM configuration using Timer 2 to control LED brightness through duty cycle variation.

UART CONFIGURATION



UART configuration for serial communication at 115200 baud rate for real-time data monitoring.

EXPECTED UART OUTPUT



Expected UART output showing the average ADC value and warning message when the threshold is exceeded.